

ARDOR / INDEPENDENT FUNCTION MODULE

Stereo headphones / LM386 / relay mute

T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

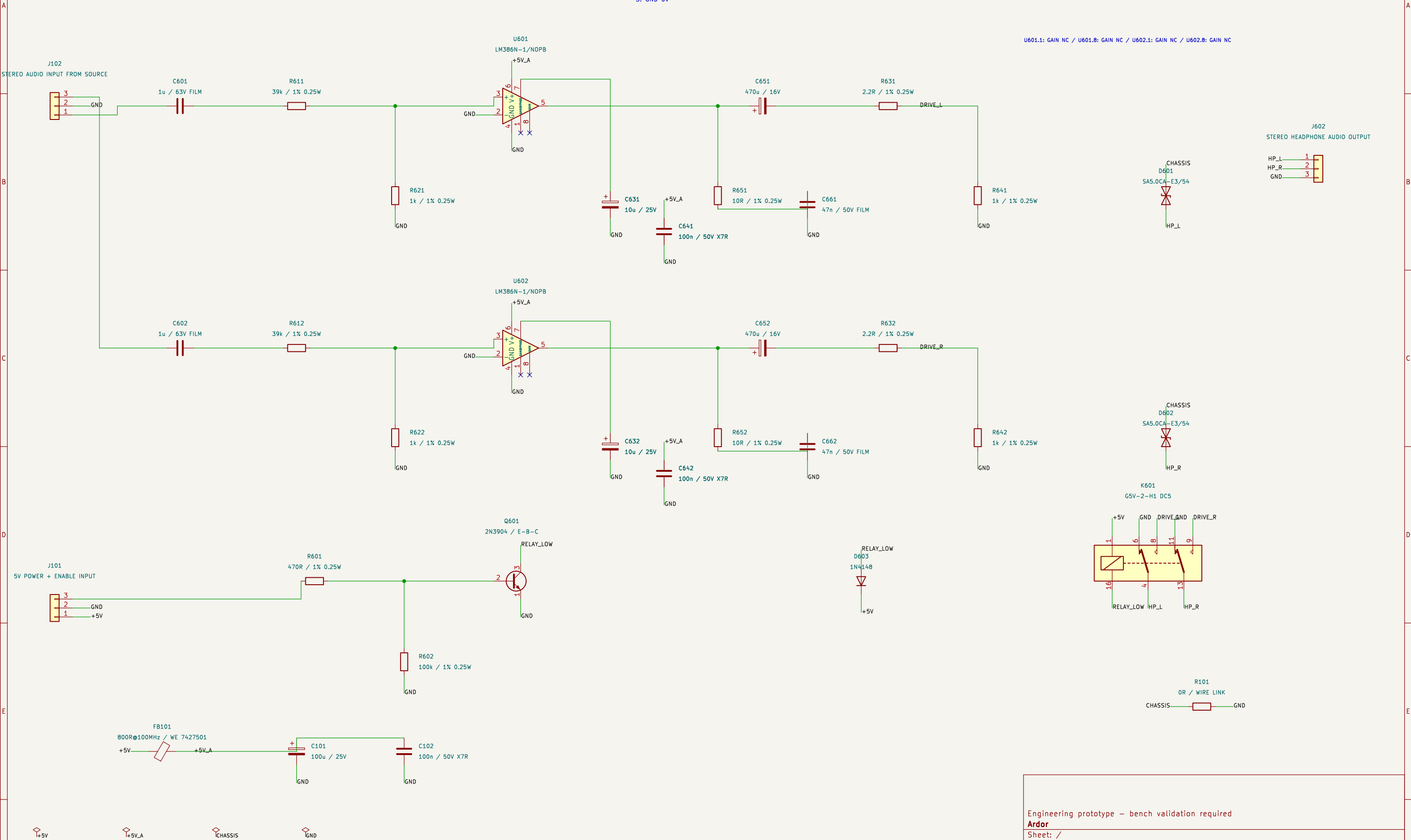
PIN GUIDE: IN = Into this board / OUT = out / I/O = both / GND = 0 V

J101 / PWR + ENABLE IN
1: 5V IN 2: GND 0V
3: ENABLE IN

J602 / HEADPHONE OUT
1: LEFT OUT 2: RIGHT OUT
3: GND 0V

J102 / AUDIO IN
1: LEFT IN 2: GND 0V
3: RIGHT IN

U601.1: GAIN NC / U601.8: GAIN NC / U602.1: GAIN NC / U602.8: GAIN NC



Engineering prototype – bench validation required

Ardor

Sheet: /
File: Ardor_Headphones_THT.kicad_sch

Title: Stereo headphones / LM386 / relay mute

Size: A3 Date: 2026-10-08

Rev: T1

KiCad E.D.A. 9.0.2

Id: 1/1

5V / 150mA budget. 32-300 ohm headphones; Input <=1Vrms. Begin at low host volume.
LM386 pins 1/8 OPEN: gain 20. 39k/1k divider gives -0.49 unloaded system gain.
470u output caps: + toward amplifier. Relay OFF grounds both panel channels.
HP_ENABLE: 3.3V HIGH sources -5mA; wait >=5s after stable power/audio before HIGH.
No charge pump. DIY audio tradeoff: noise/distortion must be measured; not the M1 hi-fi IC.